MIL-C-24643/49B
3 October 1986
SUPERSEDING
MIL-C-0024643/49A(SH)
8 November 1985
MIL-C-24643/49
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, THREE CONDUCTOR, TYPE LSTNW (INCLUDING VARIATION LSTNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSTNW (unarmored) and LSTNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. (See table I for size).
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene or ethylene propylene rubber insulation, colored black. (See table I for thickness.) Standard identification code applied by method 1.
Fourth - Three conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LSTNW

Sixth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:15Z
Check the source to verify that this is the current version before use.

Construction variant for type LSTNWA

Sixth - Cross-linked polyolefin jacket. (See table I for thickness).
Seventh - Braided metal armor and paint.

TABLE I. Construction details.

Military 1/ part no. M24643/49	Size 2/	AWG conductor size	Insulation thickness nominal (inch)	Cable jacket thickness nominal (inch)	Overall diameter (inches)			
					Type LSTNW		Type LSTNWA	
					min	max	min	max
-01UN	3	16(class B)	0.030	0.040	0.380	0.411	0.430	0.461
-02UN	4	14(class B)	.030	.040	.415	.449	.465	.499
-03UN	9	10(class B)	.030	.050	.578	.625	.628	.675
-04UN	14	9(class B)	.045	.050	.620	.670	.670	.720
-05UN	23	7(class B)	.045	.055	.703	.760	.753	.810
-06UN	50	3(class C)	.045	.060	.869	.969	.919	1.019
-07UN	75	1(class C)	.055	.070	1.048	1.134	1.098	1.184
-08UN	100	0(class D)	.055	.075	1.171	1.266	1.221	1.316
-09UN	150	000(class D)	.055	.075	1.401	1.515	1.451	1.565

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSTNW and LSTNWA.

TABLE II. Electrical details.

Size 1/	AWG conductor size	Conductor resistance per 1000 feet at 25°C (max) (ohms)	Insulation resistance/ 1000 feet (minimum) (megohms)	Voltage withstand (minimum) (volts)	Cold bending mandrel diameter (maximum) (inches)	Accelerated service loading current (amperes)
3	16(class B)	4.3	6000	4000	3	---
4	14(class B)	2.55	5000	4000	4	---
9	10(class B)	1.08	3800	4000	5	---
14	9(class B)	0.859	4400	5500	5	90
23	7(class B)	.543	3700	5500	7	115
50	3(class C)	.21	2600	5500	9	185
75	1(class C)	.134	2600	7000	10	250
100	0(class D)	.106	2400	7000	12	295
150	000(class D)	.0674	1800	7000	14	400

1/ Type LSTNW and LSTNWA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:15Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	(see table II)
Voltage withstand - volts, root mean square, minimum....	(see table II)
Insulation resistance - megohms/1000 feet, minimum.....	(see table II)
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - (See table II for mandrel diameter).....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Cross-linked polyethylene	
Insulation	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Ethylene propylene rubber	
Tensile strength - lb/in, minimum.....	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimu.....	80
Elongation - percent of unaged, minimum.....	80
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, unaged, minimum	75
Elongation - percent of unaged, minimum.....	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:15Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	25
Permanence of printing (jacket) type LSTNW only - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to materials, construction and coverage, type LSTNWA only	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cold working (cable) (minus 20 ± 2°C)	No damage
Accelerated service (cable) (sizes 14 and larger)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTNW-3	2500
LSTNW-4, 9, 14	2000
LSTNW-23	1500
LSTNW-50, 75	1000
LSTNW-100, 150	1000

Source: https://assist.dla.mil -- Downloaded: 202640-20T00:15Z
Check the source to verify that this is the current version before use.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-49)

Source: https://assist.dla.mil -- Downloaded: 20261O-20T00:15Z
Check the source to verify that this is the current version before use.